

Hospital Multistate Data Preparation

Setup

```
library(here)
library(tidyverse)
library(data.table)
library(arrow)
```

```
df <- read_parquet(here("data", "hosp_long_complete.parquet"))
```

Code Y

```
df <- df |>
  select(-na_est) |>
  mutate(
    y = case_when(
      death == 1 ~ 5,
      unplanned_stay ~ 4,
      er_visit ~ 3,
      in_hospital ~ 2,
      TRUE ~ 1
    )
  ) |>
  relocate(pat_id, tx, y)
```

Add lag + reduce to weeks for simulations

We will reduce the granularity of time to weeks for the simulations.

```
# Convert time-scale to weeks
# Day zero is counted as week 0 (baseline), Day 1 counts towards week 1.
df_weeks <- df |>
  mutate(
    week = ceiling(day / 7),
    surv_weeks = ceiling(survival_time/7)
  ) |>
  filter(week > -2) |>
  group_by(pat_id, week) |>
  arrange(desc(y)) |>
  slice_head(n = 1) |>
  arrange(pat_id, week) |>
  group_by(pat_id) |>
  mutate(
    yprev = case_when(
      TRUE ~ lag(y, n = 1, order_by = week)
    ),
    yprev = factor(yprev, levels = 1:4, ordered = FALSE), #death = 5 doesn't exist
    in_yprev
```

```

    gap = 1) |>
  slice(-1) |>
  ungroup() |>
  select(-day) |>
  relocate(pat_id, tx, y, yprev, week)

```

Create data set with ypprev (2 weeks ago) for second order model

```

df_ypprev <- df_weeks |>
  group_by(pat_id) |>
  mutate(
    ypprev = case_when(
      TRUE ~ lag(yprev, n = 1, order_by = week)
    ),
    ypprev = factor(ypprev, levels = 1:4, ordered = FALSE)) |>
  slice(-1) |>
  ungroup() |>
  relocate(pat_id, tx, y, yprev, ypprev)

df_weeks <- df_weeks |>
  filter(week > 0)

```

Data daily

```

df_daily <- df |>
  group_by(pat_id, day) |>
  arrange(desc(y)) |>
  slice_head(n = 1) |>
  arrange(pat_id, day) |>
  group_by(pat_id) |>
  mutate(
    yprev = case_when(
      TRUE ~ lag(y, n = 1, order_by = day)
    ),
    yprev = factor(yprev, levels = 1:4, ordered = FALSE),
    ypprev = case_when(
      TRUE ~ lag(yprev, n = 1, order_by = day)
    ),
    ypprev = factor(ypprev, levels = 1:4, ordered = FALSE),
    gap = 1) |>
  slice(-1) |>
  ungroup() |>
  filter(day > 0) |>
  relocate(pat_id, day, tx, y, yprev, ypprev)

```

Export

```

write_parquet(df_weeks, sink = here("data", "hosp_long_weeks.parquet"))
write_parquet(df_ypprev, sink = here("data", "hosp_long_weeks_two_lags.parquet"))
write_parquet(df_daily, sink = here("data", "hosp_long_daily.parquet"))

```